



Codec-Based Deepfake Source Tracing via Neural Audio Codec Taxonomy

Xuanjun Chen^{1*}, I-Ming Lin^{2*}, Lin Zhang⁴, Jiawei Du², Haibin Wu^{5†}
Hung-yi Lee^{1,3}, Jyh-Shing Roger Jang²

¹Graduate Institute of Communication Engineering, National Taiwan University

²Department of Computer Science and Information Engineering, National Taiwan University

³Department of Electrical Engineering, National Taiwan University

⁴Speech@FIT, Brno University of Technology ⁵Independent Researcher

Abstract

Recent advances in neural audio codec-based speech generation (CoSG) models have produced remarkably realistic audio deepfakes. We refer to deepfake speech generated by CoSG systems as codec-based deepfake, or CodecFake. Although existing anti-spoofing research on CodecFake predominantly focuses on verifying the authenticity of audio samples, almost no attention was given to tracing the CoSG used in generating these deepfakes. In CodecFake generation, processes such as speech-to-unit encoding, discrete unit modeling, and unit-to-speech decoding are fundamentally based on neural audio codecs. Motivated by this, we introduce source tracing for CodecFake via neural audio codec taxonomy, which dissects neural audio codecs to trace CoSG. Our experimental results on the *CodecFake+* dataset provide promising initial evidence for the feasibility of CodecFake source tracing while also highlighting several challenges that warrant further investigation.

Index Terms: Anti-spoofing, source tracing, audio deepfake detection, neural audio codec, explainability

1. Introduction

Deepfake detection distinguishes bona fide audio (spoken by humans) from deepfakes (generated by machines). Numerous challenges and datasets [1,2] have been created to drive research in this field, e.g., ASVspoof [3–7] and ADD [8,9]. Beyond detecting deepfake speech, tracing its source has recently gained attention. Source tracing can provide valuable cues to identify the origin or creator of deepfake audio, which is particularly useful in out-of-domain deepfake detection. The track 3 of ADD 2023 [9] focused on this topic to recognize the algorithms of the deepfake utterances. In Interspeech 2025, a special session¹ is being held to encourage researchers to develop source tracing techniques for deepfake speech.

The most straightforward way of source tracing [10] is tracing deepfake speech back to its originating attack model ID [9]. To improve generalization, researchers start working more flexibly, known as attribute-based source tracing [11–14]. Instead of using deepfake system IDs as categories, it categorizes deepfake systems based on their attributes, given that some deepfake systems share common components. This enables better generalization to trace spoofing algorithms not seen during training but composed of known building blocks [14]. Guided by well-defined attributes in conventional text-to-speech (TTS) and voice conversion (VC) systems, researchers in source tracing pay their attention to classifying attributes: input types [14], acoustic models [12,14], speaker representation [13], and vocoders [11–14], etc.

However, all the aforementioned studies focus on source tracing for conventional TTS/VC, whereas tracing sources in the emerging codec-based speech generation (CoSG) models remains largely unexplored. For the new CodecFake [15–17], source tracing was preliminarily explored in [18]. They treat the source tracing task as a closed-set classification problem for category deepfake systems ID. Although they introduced an additional binary out-of-distribution classification, it cannot provide deeper insights when encountering multiple unseen CoSG systems. Overall, source tracing for CodecFake remains limited and urgently requires exploration.

Recently, *CodecFake+* [17], a large-scale CodecFake database, was proposed. It covers most currently available neural audio codecs and CoSG systems. The dataset and its well-structured taxonomy serve as valuable reference guides for source tracing. Based on *CodecFake+*, we defined taxonomy-based source tracing tasks for CodecFake speech, including vector quantization classification, auxiliary objective classification, and decoder type classification. Additionally, we explored different training strategies and taxonomy-guided balanced training data selection. Our experiments show that a multi-task framework incorporating the three classification tasks enhances source tracing performance. Additionally, selecting balanced training data guided by auxiliary objectives shows potential for improving the generalization of source tracing. We will release our entire codebase to facilitate further research².

2. Neural Audio Codec Taxonomy

Neural audio codecs [19–22] have emerged as not only a core technology to compress and transmit audio signals efficiently, but also a core component for high-quality speech generation systems. Yet, unlike traditional TTS/VC pipelines with clearly delineated acoustic models and vocoders, codec-based speech generation systems often lack a standardized, transparent structure. This obstacle complicates detecting and tracing codec-based deepfake audio to its true source. A recent work proposed a large-scale *CodecFake+* with a comprehensively defined taxonomy [17], such as vector quantization, auxiliary objectives, and decoder types. From this, we gain a structured lens to understand and categorize various codec methods. The taxonomy is categorized into three groupings as follows:

2.1. Vector Quantization (VQ)

Vector quantization methods aim to map frame-wise audio features to discrete tokens. The VQ grouping categorization can have three sub-categories:

- **Multi-codebook Vector Quantization (Mvq):** Multiple

*Equal Contribution. † Corresponding Author

¹<https://deepfake-total.com/sourcetracing>

²<https://github.com/ResponsibleGenAI/CodecFake-Source-Tracing>

quantizers are applied sequentially to refine the residuals of previous ones.

- **Single-codebook Vector Quantization (Svq)**: Uses a single codebook to encode fewer tokens, reducing token usage.
- **Scalar-codebook Vector Quantization (Seq)**: Mapping complex signals into a compact latent space based on a scalar codebook.

2.2. Auxiliary Objectives (AUX)

Auxiliary objectives provide additional supervision to guide the training of neural audio codecs. Three sub-categories are:

- **Semantic Distillation (Sem)**: Enhances codec representation with semantic information.
- **Disentanglement (Disent)**: Separates speech attributes (e.g., content, prosody, timbre) into distinct representations.
- **None**: refer to no auxiliary Objectives.

2.3. Decoder Types (DEC)

The decoder aims to reconstruct waveforms from discrete units, with upsampling being a critical operation. There are two main methods based on the processing domain:

- **Time Decoder (Time)**: Uses transposed convolution to up-sample codec representations into raw audio waveforms.
- **Frequency Decoder (Freq)**: Uses Inverse Short-Time Fourier Transform to upsample decoded features.

In summary, the taxonomy defined above underlie nearly all neural audio codec designs. Thus, they serve as robust indicators of the “fingerprint” of a neural audio codec, enabling the tracing of CoSG systems based on their codec components, even when applied to out-of-domain CoSG systems.

3. Codec-Based Deepfake Source Tracing

3.1. Task Definition

Building on the neural audio codec taxonomy, we define three multi-class classification tasks for CodecFake source tracing:

- **Vector Quantization Classification (VQ Task)**: Identifies the codec’s vector quantization approach of a given speech.
- **Auxiliary Objective Classification (AUX Task)**: Classifies the codec’s auxiliary objective of a given speech.
- **Decoder Type Classification (DEC Task)**: Classifies the codec’s decoder type of a given speech.

Additionally, we include a traditional anti-spoof detection task:

- **Binary Spoof Detection (BIN Task)**: Determines whether a given audio is bona fide or spoofed speech.

A detailed description of the labels for each multi-class classification task is provided in Figure 1.

3.2. Multi-Task Training for Source Tracing

As shown in Figure 1, the waveform is first fed into a front-end to convert to a hidden presentation. It is then processed by four independent backends, each having a similar structure but producing outputs in different dimensions for its corresponding task, as introduced in the previous subsection. During training, the output score from each backend is used to compute its corresponding loss. Then, the total loss is obtained by a weighted sum as follows:

$$l_{\text{total}} = \lambda_1 l_{\text{BIN}} + \lambda_2 l_{\text{VQ}} + \lambda_3 l_{\text{AUX}} + \lambda_4 l_{\text{DEC}}. \quad (1)$$

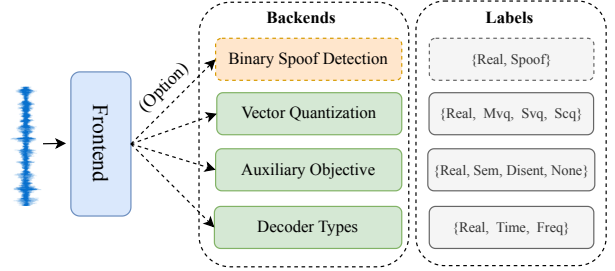


Figure 1: A multi-task source tracing framework.

Here, $l_{\text{BIN}}, l_{\text{VQ}}, l_{\text{AUX}}, l_{\text{DEC}}$ represent the losses for binary spoof detection, vector quantization classification, auxiliary objective classification, and decoder type classification, respectively. Let $[\lambda_1, \lambda_2, \lambda_3, \lambda_4]$ denote the weights for each subtask. When a subtask is added during training, its corresponding λ_i is included; otherwise, it is omitted. The study explores different training configurations, detailed in Section 4, and summarized in Table 2.

3.3. Inference for Source Tracing and Anti-Spoofing

For source tracing, each multi-class backend predicts the most probable class based on the output probabilities. For spoof detection, if the binary backend is included, its score is used directly; otherwise, the bona fide probabilities from multiple backends are multiplied, and a root transformation inspired by prior work [13] is applied. These two inference configurations are M1 and M2, respectively, as summarized in Table 2.

Since M2 was not directly optimized for binary spoof detection, we use the cubic root of the probability product as a heuristic, assuming that the VQ, AUX, and DEC tasks provide complementary cues. Based on this heuristic, when training with only the three multi-class source tracing tasks (i.e., VQ, AUX, and DEC tasks), the inference process computes the cubic root of the product of the three bona fide probabilities:

$$P^{\text{bonafide}} = \sqrt[3]{P_{\text{VQ}}^{\text{bonafide}} \times P_{\text{AUX}}^{\text{bonafide}} \times P_{\text{DEC}}^{\text{bonafide}}}, \quad (2)$$

where $P_{\text{task}}^{\text{bonafide}}$ denotes the bona fide probability predicted by the corresponding backend.

4. Experimental Setup

We investigate source tracing in *CodecFake+* [17] dataset, which comprises two subsets: CoRS (speech resynthesized by pre-trained neural audio codec models) and CoSG (speech generated by codec-based speech generation models). The original CoRS includes 31 codecs used to resynthesize spoofed audio from VCTK [23]. The full training dataset consists of $31 \times 42,965$ spoofed samples. However, due to limited computing resources, we performed taxonomy-guided balanced sampling following *CodecFake+*, selecting part of the spoofed samples to maximize data diversity while maintaining efficiency. Rather than focusing on CodecFake detection as in *CodecFake+*, our study targets source tracing. The final training dataset consists of 42,965 bona fide and 42,965 spoofed audio samples. The CoRS and CoSG test data are collected from 17 CoSG models from the *CodecFake+* dataset, primarily sourced from official demo pages. This setup better reflects real-world scenarios where spoofed data is generated using unknown systems. Table 1 summarizes the dataset statistics for source tracing.

Table 1: Summary of Source Tracing Dataset Based on CodecFake+ [17]. The BIN represents the overall CodecFake dataset statistics used in this paper, while Spoof (VQ, AUX, DEC) provides detailed statistics of the spoofed data from a taxonomy perspective.

Dataset	Sampling Strategies	BIN		Spoof (VQ)			Spoof (AUX)			Spoof (DEC)		#. of Codec
		Bona fide	Spoof	Mvq	Svq	Scq	None	Sem	Disent	Time	Freq	
CoRS Train	VQ balanced sampling	42,965	42,965	14,322	14,322	14,321	32,529	4,590	5,846	23,245	19,720	31
CoRS Validation		735	735	245	245	245	559	72	104	403	332	31
CoRS Train	AUX balanced sampling	42,965	42,965	36,152	5,278	1,535	14,322	14,321	14,322	28,445	14,520	31
CoRS Validation		735	735	624	80	31	245	245	245	469	266	31
CoRS Train	DEC balanced sampling	42,965	42,965	34,156	5,926	2,883	25,698	10,941	6,326	21,483	21,482	31
CoRS Validation		735	735	580	90	65	435	179	121	368	367	31
CoRS Test	N/A	755	22,650	18,120	3,020	1,510	12,835	6,040	3,775	13,590	9,060	31
CoSG Test (kn. codec)		259	318	249	6	63	275	5	38	240	78	6
CoSG Test (All)		866	930	848	20	62	873	5	52	340	590	9

Table 2: Training Configurations. The S^* notation represents single-task training. For example, $S1$ refers to two separate models trained independently on BIN and VQ tasks.

Model	Dataset	BIN	VQ	AUX	DEC	Initial Weight W
S1	VQ balance	✓				(1.0)
	VQ balance		✓			(1.0)
S2	AUX balance	✓				(1.0)
	AUX balance			✓		(1.0)
S3	DEC balance	✓				(1.0)
	DEC balance				✓	(1.0)
D1	VQ balance	✓	✓			(0.5, 0.5)
D2	AUX balance	✓		✓		(0.5, 0.5)
D3	DEC balance	✓			✓	(0.5, 0.5)
M1	VQ / AUX / DEC	✓	✓	✓	✓	(0.25, 0.25, 0.25, 0.25)
M2	VQ / AUX / DEC		✓	✓	✓	(0.3, 0.4, 0.3)

We use Wav2Vec2-AASIST [24]³ as the model backbone, with modifications to fit multi-task learning as shown in Figure 1. The pre-trained Wav2Vec 2.0⁴ [25] and RawNet2 [26] Encoder are shared across multiple tasks. All inputs are raw waveforms of approximately 4 seconds at a sample rate of 16 kHz, with RawBoost [27] applied for data augmentation. We train all models on an NVIDIA RTX 4070 GPU with a batch size of 8. The initial learning rate is 1×10^{-6} , and weight decay is 1×10^{-4} . Cross-entropy losses are used for all backends.

Table 2 summarizes the training configurations, where S represents Single-task Learning, D represents Dual-task Learning with BIN and one out of VQ/AUX/DEC, and M represents Multi-task learning. We use the Equal Error Rate (EER%) for deepfake detection as the evaluation metric, where lower values indicate better performance. We use the sample-weighted F1 score for three source-tracing sub-tasks to assess the performance, where higher values indicate better performance.

5. Experiment Results

5.1. Results of Source Tracing

We present performance across three source tracing tasks (defined in Section 3.1), with results in Table 3. Performance on spoof detection and source tracing is included to investigate their complementary relationship. Each sub-table corresponds to models trained using the respective balanced sampling strate-

Table 3: Results of evaluation on different source tracing tasks.

(a) Vector Quantization Classification						
Model	CoRS		CoSG (kn. codec)		CoSG (All)	
(in Table 2)	EER ↓	F1 ↑	EER ↓	F1 ↑	EER ↓	F1 ↑
Random	50.28	32.26	50.82	29.21	49.97	30.79
S1	2.78	97.39	21.66	42.94	21.93	35.26
D1	2.62	96.25	24.27	45.55	15.58	38.78
M1	2.09	97.15	30.85	38.54	16.97	35.08
M2	2.38	96.58	30.15	45.77	36.95	38.48
(b) Auxiliary Objective Classification						
Model	CoRS		CoSG (kn. codec)		CoSG (All)	
(in Table 2)	EER ↓	F1 ↑	EER ↓	F1 ↑	EER ↓	F1 ↑
Random	50.28	28.78	50.82	30.94	49.97	31.04
S2	2.77	97.15	22.71	32.80	15.02	19.44
D2	1.58	97.46	27.74	34.22	16.86	19.18
M1	2.12	95.94	33.62	33.66	19.64	15.92
M2	1.17	96.90	30.50	41.22	21.03	28.18
(c) Decoder Types Classification						
Model	CoRS		CoSG (kn. codec)		CoSG (All)	
(in Table 2)	EER ↓	F1 ↑	EER ↓	F1 ↑	EER ↓	F1 ↑
Random	50.28	38.97	50.82	36.89	49.97	34.37
S3	2.38	97.96	24.78	46.73	11.91	27.26
D3	1.48	96.67	29.29	47.94	34.89	16.56
M1	1.31	97.42	32.41	38.26	20.09	10.86
M2	1.05	97.97	36.23	38.71	33.39	13.42

gies and evaluated on the corresponding tasks. Each row represents a different model. For S^* models, performance is reported from two separate models: EER is measured using the one trained on BIN task, while F1 is measured using the one trained on VQ/AUX/DEC. From D^* to M2, each row corresponds to a single model evaluated across different datasets. We also include the random guess as the baseline. Each column represents a different evaluation subset.

CoRS evaluation set shows the performance when training and evaluation data are re-synthesized using the same neural audio codecs. For deepfake detection, EER decreases when introducing more backend branches, from single-task and dual-task setups to multi-task learning, boosting overall performance. For source tracing, across the three multi-class classification tasks, all models achieve F1 scores of roughly 96%–97%.

CoSG (kn. codec) set presents performance when evaluating only the CoSG data that involves the known codec. In this set, the best EERs for VQ, AUX, and DEC tasks reach 21.66%, 22.71%, and 24.78%, respectively. These are about an order of

³https://github.com/TakHemlata/SSL_Anti-spoofing

⁴<https://huggingface.co/facebook/wav2vec2-xls-r-300m>

magnitude worse than the CoRS evaluation. Multi-task learning does not appear to help deepfake detection here, though it improves VQ and AUX source tracing. For DEC source tracing, dual-task learning achieves the highest F1 score at 47.94%.

The CoSG (All) set comprises all CoSG evaluation data. Compared to CoSG (kn. codec), the best deepfake detection EER improves to 15.58%, 15.02%, and 11.91% across the three sampled datasets. For source tracing, the multi-task M2 model outperforms or approaches dual-task and single-task approaches, though the decoder-types task achieves only a 13.42% F1 score. Source-tracing tasks generally benefit deepfake detection only within the CoRS evaluation set. Meanwhile, the M2 model delivers competitive performance with best model in VQ task and AUX task for both CoSG (kn. codec) and CoSG (All).

Although multi-task learning benefits VQ and AUX source tracing tasks, model performance declines as the mismatch increases from CoRS to CoSG (kn. codec) to CoSG (All). This trend can be explained by the codec-based deepfake generation process, which involves two steps: (1) a neural codec encodes speech into discrete units, and (2) a generative model processes these units before a codec decoder generate speech. Based on the above context, we have two reasons to speculate: Firstly, when trained on CoRS data from a specific codec, the model performs better on CoSG (kn. codec), where only the generative modeling step differs. This suggests that generative modeling significantly impacts source tracing performance. Secondly, performance further declines in CoSG (All) due to the using of unknown codecs (e.g., Mel-VAE [28], SoundStream [29], Single-Codec [30], DAC-Vocos [31]) in the corresponding CoSG system [28, 30–32], adding another variability. Our findings highlight that generative modeling and unseen codecs influence CodecFake source tracing.

5.2. The Impact of Various Data Balance Strategies

We train the model on three taxonomy-guided balanced training sets (listed in Table 1) to examine how various data balancing strategies affect source tracing. The results are in Table 4. Although the DEC-balanced dataset-trained D3 performs slightly better on the DEC task, AUX-balanced dataset-trained models perform well among the three tasks. Notably, when trained on the AUX-balanced dataset, M2 got the best performance on both the VQ (52.07%) and AUX (41.22%) tasks, with a competitive F1 score of 46.45% on the DEC task. These results indicate that model performance is susceptible to variations in training data distributions. Our experiments suggest that balancing the training sets according to auxiliary objectives yields the strongest generalization on source tracing tasks.

5.3. Error Analysis

In the previous section, we observed that M2, trained on the AUX-balanced dataset, achieves the best generalization performance on source tracing. To further investigate, we compared the confusion matrices of model M1 and M2 (both trained on the AUX-balanced dataset) evaluation on different source tracing tasks as shown in Figure 2. M1 frequently misclassifies bona fide speech, whereas M2 correctly identifies more bona fide samples, significantly improving overall performance.

One reason M1 underperforms M2 is its multitask approach, which must handle both bona fide/spoof classification and multi-class classification for source tracing, which can often be confusing. Meanwhile, M2 focuses solely on source tracing, leading to more consistent performance. Another factor is a domain mismatch in bona fide speech data, which leads to mis-

Table 4: Comparison of different data balance strategies. F1 scores are evaluated on CoSG (kn. codec) subset. (Darker cell color indicates better performance)

		VQ Task (F1 ↑)	AUX Task (F1 ↑)	DEC Task (F1 ↑)
Random		29.21	30.94	36.89
VQ balanced sampling	S1	42.94	N/A	N/A
	D1	45.55	N/A	N/A
	M1	38.54	30.09	38.75
	M2	45.77	33.68	39.23
AUX balanced sampling	S2	N/A	32.80	N/A
	D2	N/A	34.22	N/A
	M1	42.25	33.66	40.01
	M2	52.07	41.22	46.45
DEC balanced sampling	S3	N/A	N/A	46.73
	D3	N/A	N/A	47.94
	M1	44.91	27.19	38.26
	M2	41.11	31.94	38.71

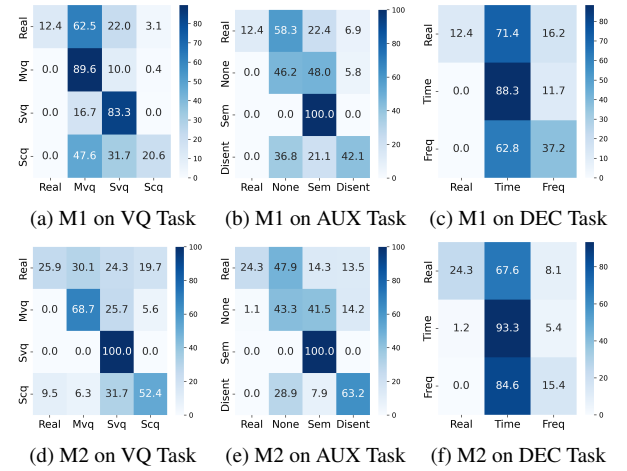


Figure 2: Confusion matrices of M1 and M2 on CoSG (kn. codec) subset, row-normalized by true labels, with predictions on the horizontal axis and true labels on the vertical axis.

classifying bona fide speech as a certain category in codec taxonomy and diminishes overall generalization. Another concurrent study [18] also noted that unseen real speech significantly reduced source tracing performance in codec identification.

6. Conclusion

This paper defines source tracing for deepfakes generated by codec-based speech generation models. We introduce three taxonomy-guided multi-class classification tasks: vector quantization, auxiliary objectives, and decoder type. Building on our defined source tracing framework, we compare different training strategies, including single-task, dual-task, and multi-task learning. We also explore taxonomy-guided data balance selection. Results on the *CodecFake+* dataset show that source tracing models are susceptible to training data distributions.

As general-purpose audio-language models become central to speech understanding [33], many rely on them as default tools. Yet even the leading models like GPT-4o may decline deepfake detection requests due to built-in safety constraints [34]. In contrast, leveraging a neural audio codecs taxonomy, our source-tracing framework enables precise analysis of codec-based speech synthesis. We anticipate that this work will yield valuable insights and inspire further research.

7. Acknowledgements

This work was partially supported by the National Science and Technology Council, Taiwan (Grant no. NSTC 112-2634-F-002-005, Advanced Technologies for Designing Trustable AI Services). We also thank the National Center for High-performance Computing (NCHC) of National Applied Research Laboratories (NARLabs) in Taiwan for providing computational and storage resources.

8. References

- [1] M. Li, Y. Ahmadiadi, and X.-P. Zhang, "A survey on speech deepfake detection," *ACM Comput. Surv.*, vol. 57, no. 7, 2025.
- [2] H. Wu, J. Kang, L. Meng, H. Meng, and H.-y. Lee, "The defender's perspective on automatic speaker verification: An overview," *arXiv preprint arXiv:2305.12804*, 2023.
- [3] Z. Wu, T. Kinnunen, N. Evans, J. Yamagishi, C. Hanilçi, M. Sahidullah, and A. Sizov, "Asvspoof 2015: the first automatic speaker verification spoofing and countermeasures challenge," in *Proc. Interspeech*, 2015.
- [4] T. Kinnunen, M. Sahidullah, H. Delgado, M. Todisco, N. Evans, J. Yamagishi, and K. A. Lee, "The asvspoof 2017 challenge: Assessing the limits of replay spoofing attack detection," in *Proc. Interspeech*, 2017.
- [5] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, H. Delgado, A. Nautsch, J. Yamagishi, N. Evans, T. H. Kinnunen, and K. A. Lee, "ASVspoof 2019: future horizons in spoofed and fake audio detection," in *Proc. Interspeech*, 2019.
- [6] X. Liu, X. Wang, M. Sahidullah, J. Patino, H. Delgado, T. Kinnunen, M. Todisco, J. Yamagishi, N. Evans, A. Nautsch, and K. A. Lee, "Asvspoof 2021: Towards spoofed and deepfake speech detection in the wild," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 31, 2023.
- [7] X. Wang, H. Delgado, H. Tak, J.-w. Jung, H.-j. Shim, M. Todisco *et al.*, "ASVspoof 5: Crowdsourced speech data, deepfakes, and adversarial attacks at scale," in *Proc. ASVspoof Workshop*, 2024.
- [8] J. Yi, R. Fu, J. Tao, S. Nie, H. Ma, C. Wang, T. Wang *et al.*, "Add 2022: the first audio deep synthesis detection challenge," in *Proc. ICASSP*, 2022.
- [9] J. Yi, C. Y. Zhang, J. Tao, C. Wang, X. Yan, Y. Ren, H. Gu, and J. Zhou, "Add 2023: Towards audio deepfake detection and analysis in the wild," *arXiv preprint arXiv:2408.04967*, 2024.
- [10] C. Borrelli, P. Bestagini, F. Antonacci, A. Sarti, and S. Tubaro, "Synthetic speech detection through short-term and long-term prediction traces," *EURASIP Journal on Information Security*, vol. 2021, 2021.
- [11] X. Yan, J. Yi, J. Tao, C. Wang, H. Ma, T. Wang, S. Wang, and R. Fu, "An initial investigation for detecting vocoder fingerprints of fake audio," *Proceedings of the 1st International Workshop on Deepfake Detection for Audio Multimedia*, 2022.
- [12] C. Y. Zhang, J. Yi, J. Tao, C. Wang, and X. Yan, "Distinguishing neural speech synthesis models through fingerprints in speech waveforms," in *China National Conference on Chinese Computational Linguistics*, 2023.
- [13] T. Zhu, X. Wang, X. Qin, and M. Li, "Source tracing: detecting voice spoofing," in *Proc. APSIPA ASC*, 2022.
- [14] N. Klein, T. Chen, H. Tak, R. Casal, and E. Khoury, "Source tracing of audio deepfake systems," in *Proc. Interspeech*, 2024.
- [15] H. Wu, Y. Tseng, and H. yi Lee, "CodecFake: Enhancing anti-spoofing models against deepfake audios from codec-based speech synthesis systems," in *Proc. Interspeech*, 2024.
- [16] Y. Xie, Y. Lu, R. Fu, Z. Wen, Z. Wang, J. Tao, X. Qi, X. Wang, Y. Liu, H. Cheng *et al.*, "The codefake dataset and countermeasures for the universally detection of deepfake audio," *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- [17] X. Chen, J. Du, H. Wu, L. Zhang, I. Lin, I. Chiu, W. Ren, Y. Tseng, Y. Tsao, J.-S. R. Jang, and H.-y. Lee, "CodecFake+: A large-scale neural audio codec-based deepfake speech dataset," *arXiv preprint arXiv:2501.08238*, 2025.
- [18] Y. Xie, X. Wang, Z. Wang, R. Fu, Z. Wen, S. Cao, L. Ma, C. Li, H. Cheng, and L. Ye, "Neural codec source tracing: Toward comprehensive attribution in open-set condition," *arXiv preprint arXiv:2501.06514*, 2025.
- [19] S. Arora, K.-W. Chang, C.-M. Chien, Y. Peng, H. Wu, Y. Adi, E. Dupoux, H.-Y. Lee, K. Livescu, and S. Watanabe, "On the landscape of spoken language models: A comprehensive survey," *arXiv preprint arXiv:2504.08528*, 2025.
- [20] H. Wu, X. Chen, Y.-C. Lin, K.-w. Chang, H.-L. Chung, A. H. Liu, and H.-y. Lee, "Towards audio language modeling-an overview," *arXiv preprint arXiv:2402.13236*, 2024.
- [21] H. Wu, H.-L. Chung, Y.-C. Lin, Y.-K. Wu, X. Chen, Y.-C. Pai *et al.*, "Codec-SUPERB: An in-depth analysis of sound codec models," in *Findings Assoc. Comput. Linguist.*, 2024.
- [22] H. Wu, X. Chen, Y.-C. Lin, K. Chang, J. Du, K.-H. Lu *et al.*, "Codec-SUPERB@SLT 2024: A lightweight benchmark for neural audio codec models," in *Proc. IEEE Spoken Lang. Technol. Workshop*, 2024.
- [23] J. Yamagishi, C. Veaux, K. MacDonald *et al.*, "Cstr vctk corpus: English multi-speaker corpus for cstr voice cloning toolkit (version 0.92)," *Univ. of Edinburgh, The Centre for Speech Technology Research (CSTR)*, 2019.
- [24] H. Tak, M. Todisco, X. Wang, J.-w. Jung, J. Yamagishi, and N. Evans, "Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation," in *Proc. Odyssey Speaker Lang. Recognit. Workshop*, 2022.
- [25] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. NeurIPS*, vol. 33, 2020.
- [26] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, "End-to-end anti-spoofing with rawnet2," in *Proc. ICASSP*, 2021.
- [27] H. Tak, M. Kamble, J. Patino, M. Todisco, and N. Evans, "Rawboost: A raw data boosting and augmentation method applied to automatic speaker verification anti-spoofing," in *Proc. ICASSP*, 2022.
- [28] J. Kim, K. Lee, S. Chung, and J. Cho, "CLaM-TTS: Improving neural codec language model for zero-shot text-to-speech," in *Proc. ICML*, 2024.
- [29] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "Soundstream: An end-to-end neural audio codec," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 30, 2021.
- [30] H. Li, L. Xue, H. Guo, X. Zhu, Y. Lv, L. Xie *et al.*, "Single-codec: Single-codebook speech codec towards high-performance speech generation," in *Proc. Interspeech*, 2024.
- [31] Y. Wang, H. Zhan, L. Liu, R. Zeng, H. Guo, J. Zheng *et al.*, "MaskGCT: Zero-shot text-to-speech with masked generative codec transformer," in *Proc. ICLR*, 2025.
- [32] K. Shen, Z. Ju, X. Tan, E. Liu, Y. Leng, L. He *et al.*, "Naturalspeech 2: Latent diffusion models are natural and zero-shot speech and singing synthesizers," in *Proc. ICLR*, 2024.
- [33] C.-Y. Huang *et al.*, "Dynamic-SUPERB phase-2: A collaboratively expanding benchmark for measuring the capabilities of spoken language models with 180 tasks," in *Proc. ICLR*, 2025.
- [34] Y.-X. Lin, C.-K. Yang, W.-C. Chen, C.-A. Li, C.-Y. Huang, X. Chen, and H.-y. Lee, "A preliminary exploration with gpt-4o voice mode," *arXiv preprint arXiv:2502.09940*, 2025.